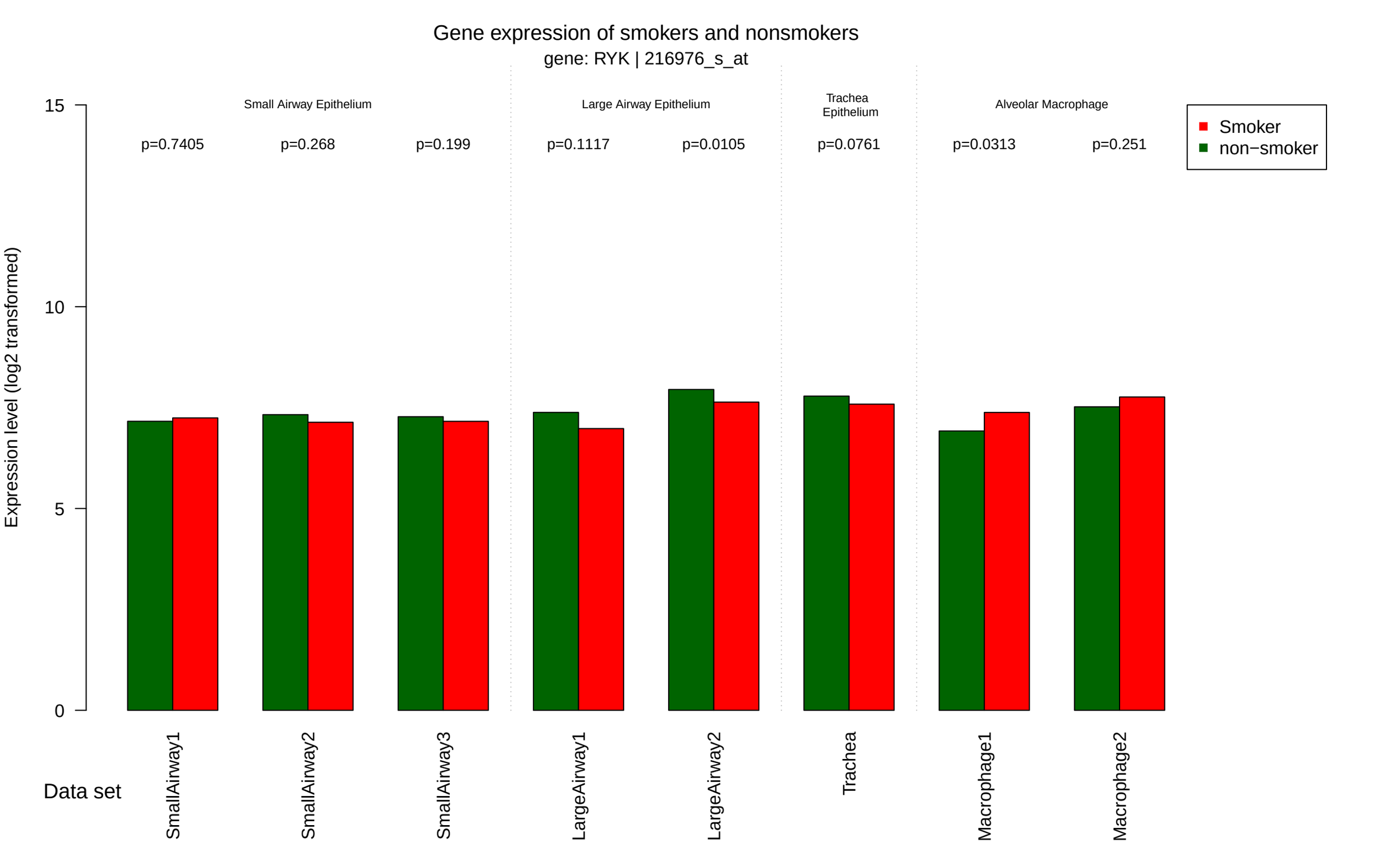
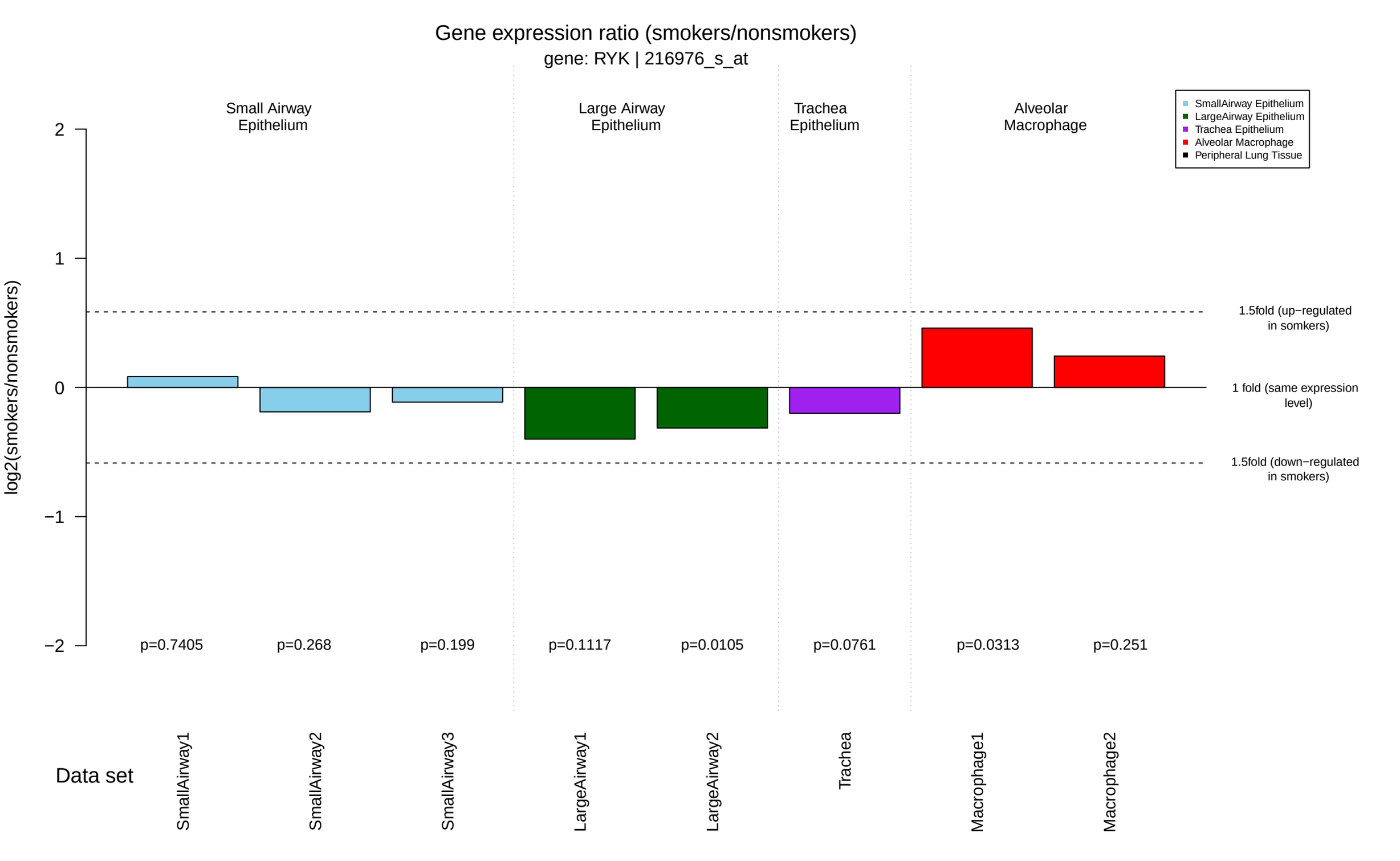
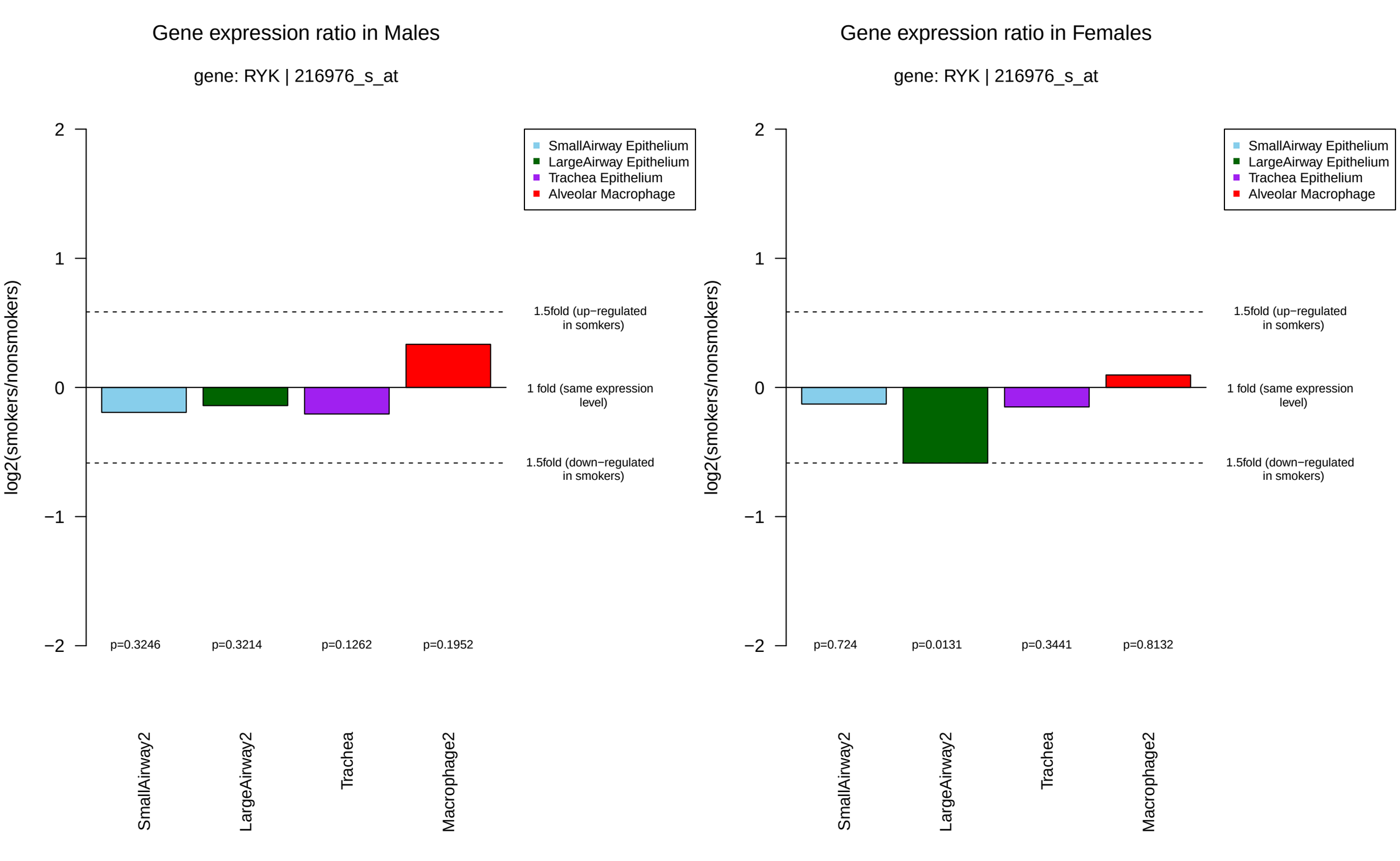
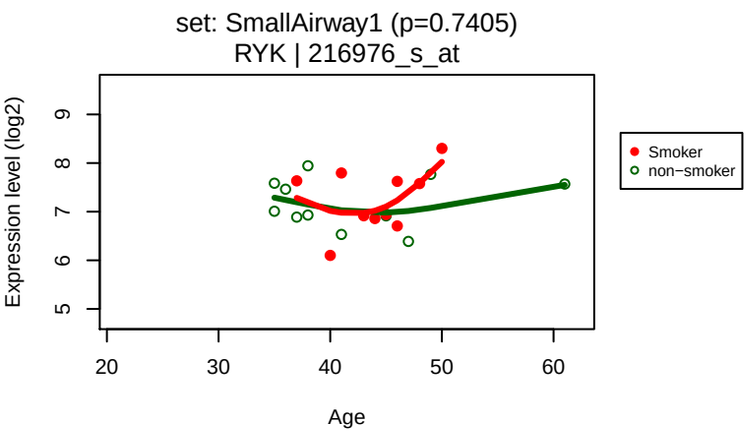
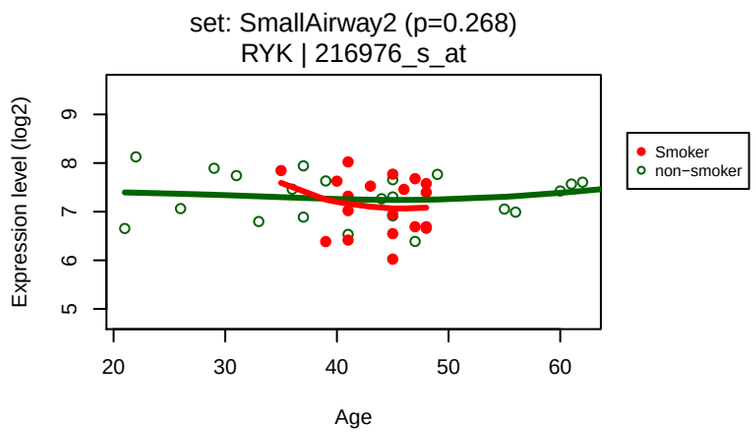
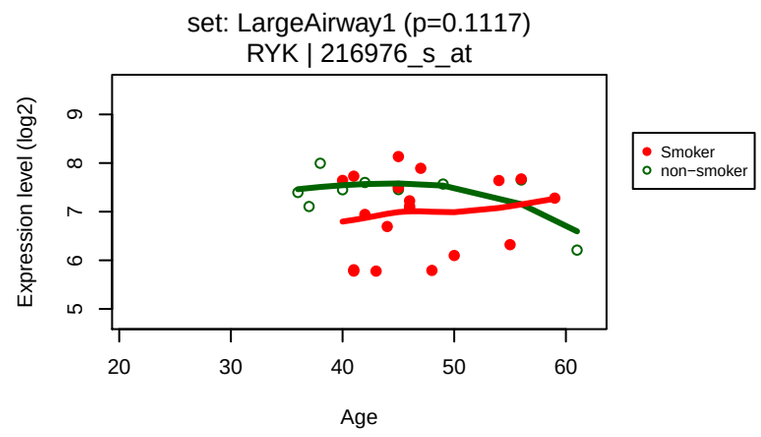

Gene expression of smokers and nonsmokers
gene: RYK | 216976_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7405
p=0.268
p=0.199
p=0.1117
p=0.0105
p=0.0761
p=0.0313
p=0.251
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

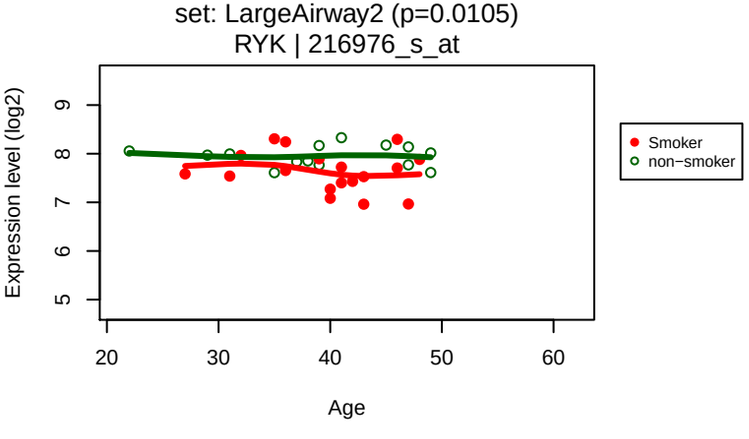

Gene expression ratio (smokers/nonsmokers)
gene: RYK | 216976_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7405
p=0.268
p=0.199
p=0.1117
p=0.0105
p=0.0761
p=0.0313
p=0.251
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

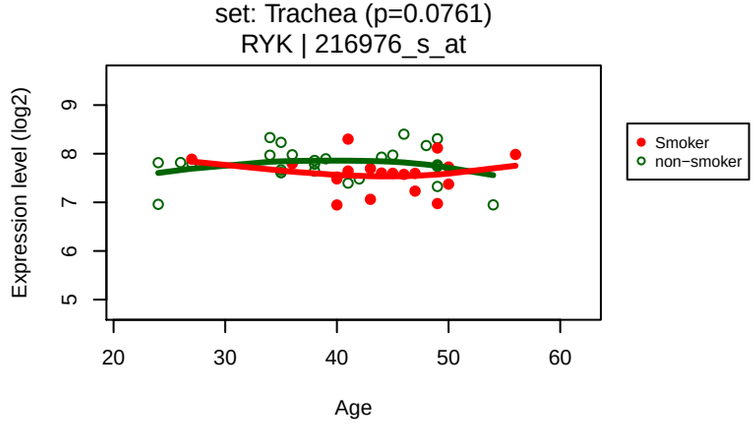

Gene expression ratio in Males
gene: RYK | 216976_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3246
p=0.3214
p=0.1262
p=0.1952
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RYK | 216976_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.724
p=0.0131
p=0.3441
p=0.8132
SmallAirway2
LargeAirway2
Trachea
Macrophage2

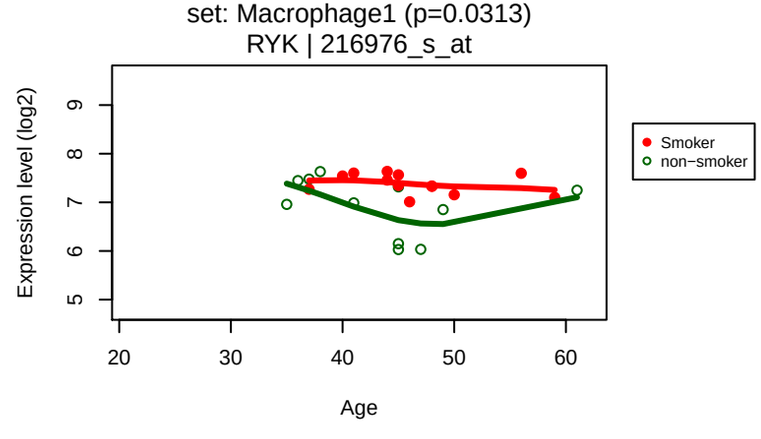

set: SmallAirway1 (p=0.7405)
RYK | 216976_s_at
Expression level (log2)
Age
Smoker
non−smoker

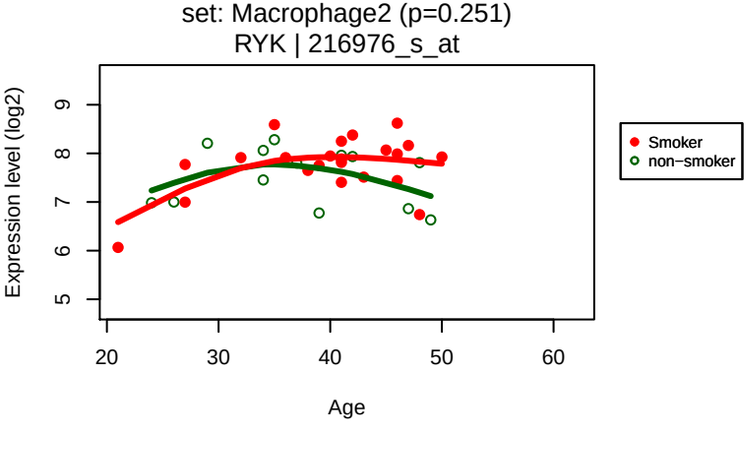

set: SmallAirway2 (p=0.268)
RYK | 216976_s_at
Expression level (log2)
Age
Smoker
non−smoker

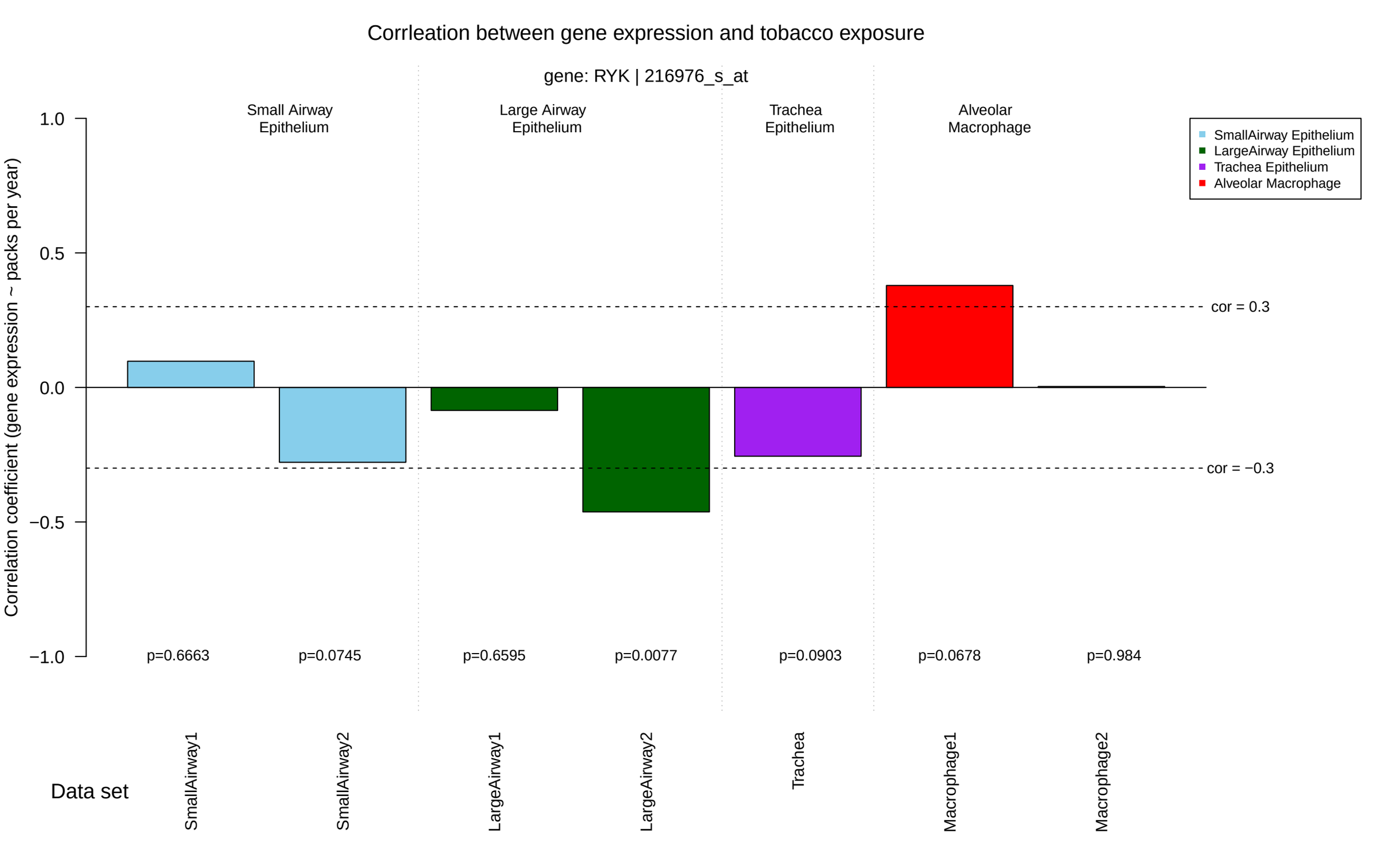

set: LargeAirway1 (p=0.1117)
RYK | 216976_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0105)
RYK | 216976_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0761)
RYK | 216976_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0313)
RYK | 216976_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.251)
RYK | 216976_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RYK | 216976_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.6663
p=0.0745
p=0.6595
p=0.0077
p=0.0903
p=0.0678
p=0.984
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2